

# Elements, Compounds & Mixtures

**Use the cue, then fade the cue.** Look at the cue card for 60 seconds. Then cover it before answering Round

1. Check the card only after you commit to an answer.

**Cue focus**

An element contains one kind of atom, a compound has elements chemically bonded in fixed combinations, and a mixture contains substances together without chemical bonding.

## Round 1 - Retrieve It

1. What makes a substance an element?

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2. What makes a substance a compound?

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3. What makes a sample a mixture?

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4. Which can usually be separated by physical methods: a compound or a mixture?

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## Round 2 - Sketch It

Create three particle sketches: one element, one compound, and one mixture. Use circles of different patterns or labels to show different atom types.



# Elements, Compounds & Mixtures - Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

## Round 3 - Apply the Relationship

1. Classify oxygen gas, O<sub>2</sub>: element, compound, or mixture?

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2. Classify carbon dioxide, CO<sub>2</sub>: element, compound, or mixture?

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3. Classify saltwater: element, compound, or mixture?

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## Round 4 - Reason from Evidence

A student says, "If two kinds of atoms are present, the sample must be a compound." Explain why that statement is incomplete.

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# Elements, Compounds & Mixtures - ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

**A****Articulate**

Explain Elements, Compounds & Mixtures in your own words. Use one example that is not printed on the cue card.

**C****Connect**

Connect particle diagrams to classification. What visual evidence would show that two substances are mixed but not bonded?

**E****Extend**

You are given iron filings and sand. Propose a physical method to separate the mixture and explain why it works.

**Confidence check** Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it

# TEACHER KEY - Elements, Compounds & Mixtures

**Remove this page before student copying.** Answers below give expected ideas, not required wording.

## Retrieve It

1. It contains only one kind of atom.
2. Two or more different elements are chemically bonded in a fixed ratio.
3. Two or more substances are physically combined but not chemically bonded.
4. A mixture.

## Sketch It - look for

Element: one atom type only. Compound: identical bonded groups containing at least two atom types. Mixture: two or more particle types together without all being chemically bonded into one repeating substance.

## Apply It

1. Element, because it contains only oxygen atoms.
2. Compound, because carbon and oxygen are chemically bonded.
3. Mixture, because salt and water are physically combined and can be separated physically.

## Reason from Evidence - look for

A mixture can also contain more than one kind of atom or molecule. A compound requires different elements to be chemically bonded in a fixed composition.

## ACE - Extend look-for

Use a magnet to remove the iron because magnetism is a physical property; the substances are not chemically bonded.

### Suggested use

Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.